

Semiconductor Memory

1. Compare NAND and NOR flash with respect to the following features.
 - Storage density
 - Random/Sequential Read
 - Random/Sequential Program
 - NAND flash has a higher storage density than NOR flash. NAND flash cells are connected in series with less wires and simpler connections, compared to NOR flash cells which allows individual cell access hence requires more connection lines. The simpler cell design for NAND flash allows greater compaction of IC layout.
 - NOR allows faster Random read as each cell can be accessed individually, compared to NAND where the entire page must be read before extracting the target byte. However, NAND's read performance improves for sequential read as access is done at page level. Similar sequential read for NOR incurs more overhead as each individual cells have to be accessed one at a time.
 - Similar to read operation, NOR has advantage with Random programming (write) as each cell can be addressed individually, compare to NAND where the entire page must be re-programmed to change one byte in the page. Similarly, Nand performs better when programming in bulk for sequential addresses due to less overhead operations and NAND block erase is much faster than NOR block erase. Recall that programming for flash requires the destination to be in erased state.

Example: NAND Block Erase (1Mbyte Block size) – 7ms



32Gb, 64Gb, 128Gb, 256Gb Asynchronous/Synchronous NAND
Electrical Specifications – Array Characteristics

Electrical Specifications – Array Characteristics**Table 43: Array Characteristics**

Parameter	Symbol	Typ	Max	Unit	Notes
Number of partial page programs	NOP	–	4	Cycles	1
ERASE BLOCK operation time	^t BERS	1.5	7	ms	
Change column setup time to data in/out or next command	^t CCS	–	200	ns	

Example: NOR Block Erase (128Kbyte) – 200ms

**512Mb: x8/x16, 3V, MT28EW Embedded Parallel NOR
Program/Erase Characteristics****Program/Erase Characteristics****Table 35: Program/Erase Characteristics**

Notes 1 and 2 apply to entire table

Parameter	Buffer Size	Byte	Word	Min	Typ	Max	Unit	Notes
Erase								
Block erase (128KB)	–	–	–	–	200	1100	ms	–
Chip erase (512Mb)	–	–	–	–	104	–	s	–

2. The SSD controller manage the translation between Logical Block Address (LBA) from the Application processor to the Physical Block Address (PBA) of the actual storage location of the data in the SSD.
- Table 2.1 shows the mapping of LBA to PBA.
 - For simplicity's sake, we can assume that these addresses at block level address, e.g. LBA block 0 (LBA0) is mapped to PBA block 9 (PBA9) in the table.
 - Table 2.2 shows the erase count of each PBA block.
 - LBA, PBA block correspond to the flash block in the flash structure, i.e. erasure is at block level.
 - A 'Y' in the 'Empty' column indicates that the Flash Block is empty.
- (a) Complete Table 2.1 for Dynamic wear leveling algorithm. You can further assume that
- Access sequence from the processor is LBA 7, 4, 3, A.
 - All access will go through the wear levelling algorithm for physical block assignment.
 - If the LBA was originally assigned to a different PBA, the PBA containing the old data will be erased.

What is the consideration to take for the erasure process of the old data?

Table 2.1	
LBA	PBA
0	9
1	X
2	X
3	5
4	X
5	4
6	6
7	7
8	X
9	X
A	0

Table 2.2		
Erase Count Table		
PBA	Erase Count	Empty
0	5	N
1	33	Y
2	15	Y
3	98	Y
4	45	N
5	11	N
6	69	N
7	78	N
8	A3	Y
9	F1	N
A	56	Y

- Evaluation of which PBA block to assign is done before any PBA block erasure. New data must be re-written to the new target PBA block before erasing the old data.

Dynamic

Table 2.1	
LBA	PBA
0	9
1	X
2	X
3	5 → A(3)
4	X → 1(2)
5	4
6	6
7	7 → 2(1),
8	X
9	X
A	0 → 5(4)

Table 2.2 (Dynamic)		
Erase Count Table		
PBA	Erase Count	Empty
0	5 → 6(4)	N → Y(4)
1	33	Y → N(2)
2	15	Y → N(1)
3	98	Y
4	45	N
5	11 → 12(3)	N → Y(3) → N(4)
6	3	N
7	78 → 79(1)	N → Y(1)
8	A3	Y
9	F1	N
A	56	Y → N(3)

- (b) In table 2.2, you can see that PBA block 6 has a very low erase count but would not be evaluated under Dynamic wear levelling algorithm. Static wear levelling attempts to relocate such flash blocks that are seldom re-written to achieve more even erase count distribution. There are many ways to implement Static wear levelling, below is one example. Go through the algorithm described and suggest enhancement to the algorithm.

- On every access, all blocks (occupied and empty) will be evaluated for block assignment.
- PBA block with the lowest erase count will be selected to store the data.

Enhancement Suggestions

- In the current method, if the chosen block to write into is already occupied, the target block will have to be erase and relocated, this will result in more erasure operation, increasing the total erase counts. To avoid this issue, separate the evaluation of occupied and empty flash blocks
 - Evaluate empty blocks to decide which flash blocks to use for active re-writing of blocks.
 - Evaluate occupied blocks periodically to re-locate occupied blocks with low erase counts, freeing these blocks for usage for subsequent access.
- Occupied blocks in part (b) above are only relocated to empty blocks to avoid unnecessary erasure process.
- Empty blocks with high erase counts can be chosen to store data which are seldom re-written into. May or may not be possible as this will need info on LBA access statistics, and provided the files associated with the LBA stays constant.

3. Figure 3.1 and 3.2 shows the read and write access timing diagram of a SRAM. It is the signal that SRAM expects or output during a Read/Write operation. When a processor interface to the SRAM, it needs to output signals conforming to what the SRAM expect and needs to read in signals SRAM generates, all within specified timings e.g. t_{OE} , t_{OHZ} etc. Describe the significance of the following parameters during Read/Write operation, and whether the parameters should be a min or max value.
- t_{OE} , t_{OHZ}
 - t_{WP} , t_{DH}

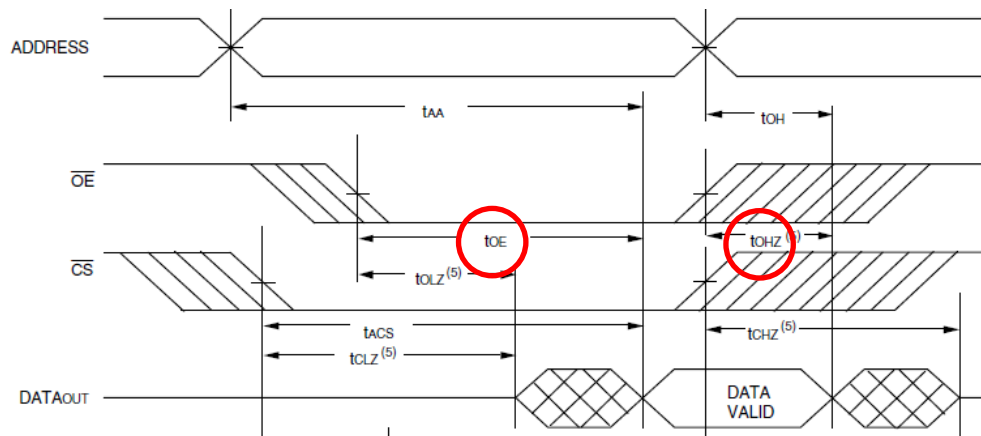


Figure 3.1: SRAM Read access timing diagram

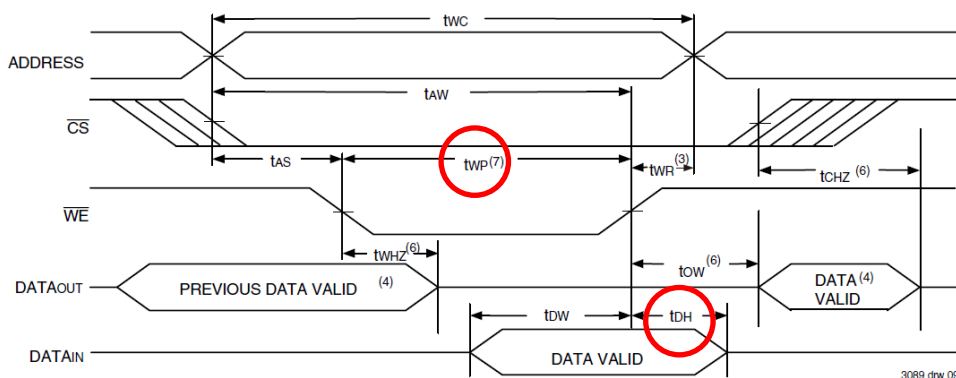


Figure 3.2: SRAM Write access timing diagram

- tOE, tOH
 - tOE is the maximum time that the SRAM needs to output the required data after its Read Strobe (OE_) is active (Logic LOW for this case).
 - tOHZ is the maximum amount of time the SRAM will hold the data on the data bus after OE_ signal is inactive.
 - The processor involved in the read operation can only perform the actual read on the data bus tOE time after asserting OE_, else the data on the bus may not be reliable.
 - Some processors require data on the data bus to be stable for a certain period after the OE_ signal is inactive. For that, ensure that the processor can support the tOHZ given in the SRAM specification, i.e. data should be read within tOHZ time after tOE is disabled.
 - tOE gives a gauge to the read access time of the SRAM, i.e. how quickly data can be read from the SRAM.
- tWP, tDH
 - tWP is the minimum time the SRAM needs after the Write Strobe (WE_) is active, in order for data on the data bus to be written into the SRAM reliably.
 - tDH is the minimum time the SRAM needs the data to be on the data bus after WE_ is inactive, in order for the SRAM to latch (read) the data reliably.
 - The processor involved in the write operation needs to
 - supply the write data on the data bus asap after asserting WE_
 - ensure WE_ is active for at least tWP time
 - hold the write data on the bus for at least tDH time after WE_ is inactive.
 - tWP and tDH give a gauge to the write access time of the SRAM.
- Note that name of these signals are vendor specific and control signals for write operation may be slightly different for different vendors.

4. Complete the timing diagrams in Figure 4.1 and 4.2 below, fill in the missing information on commands, data, addresses etc.
- Other than supplying address information, what are other functions of A10 signal?
 - What is the CAS latency of the SDRAM for Figure 4.1?
 - READ: A10=0 => No Auto-Precharge, A10=1 => Auto-Precharge
 - PRECHARGE: A10=0 => Single Precharge, A10=1 => All Banks Precharge
 - Table 6.1: CAS Latency =2.

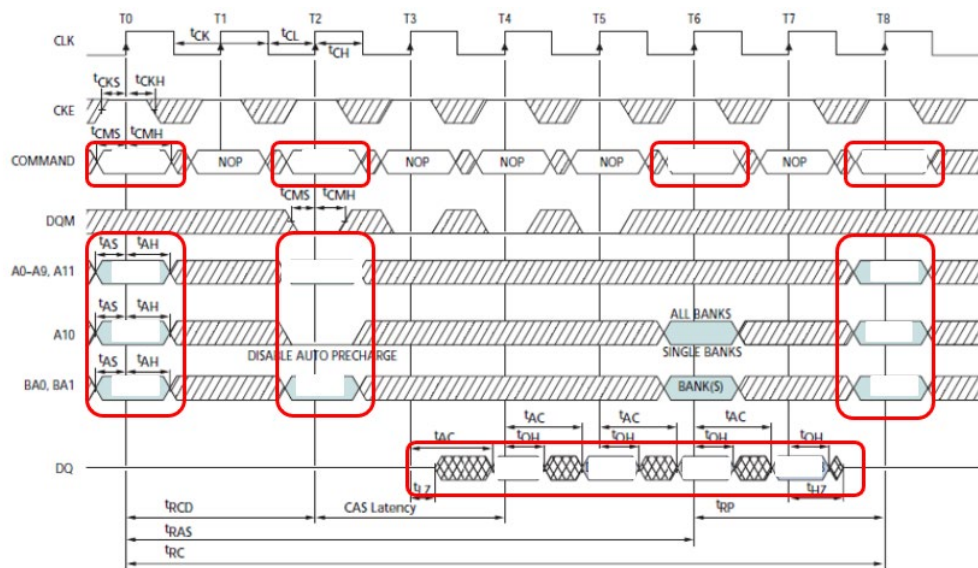
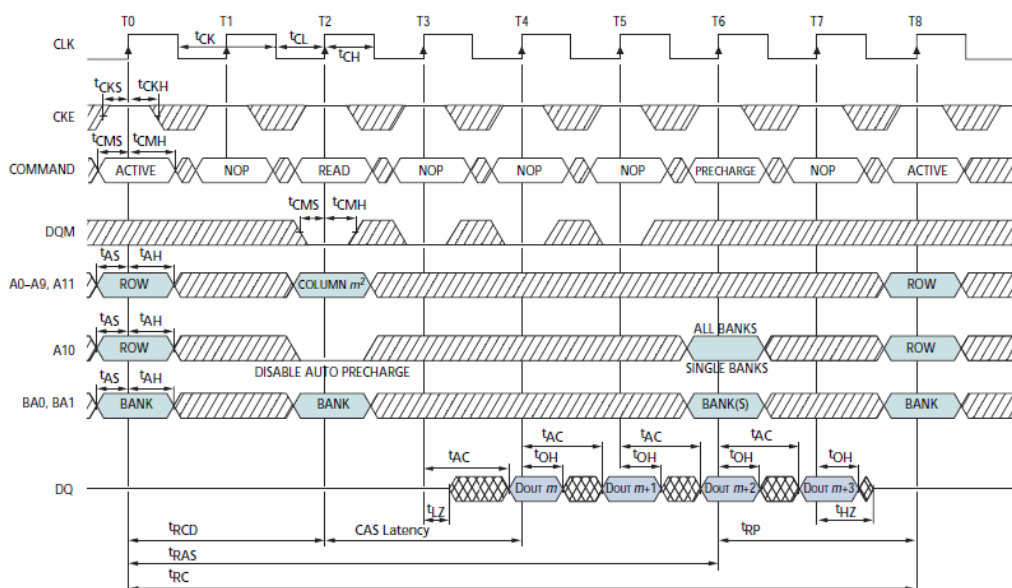


Figure 4.1: SDRAM Read without Auto-Precharge



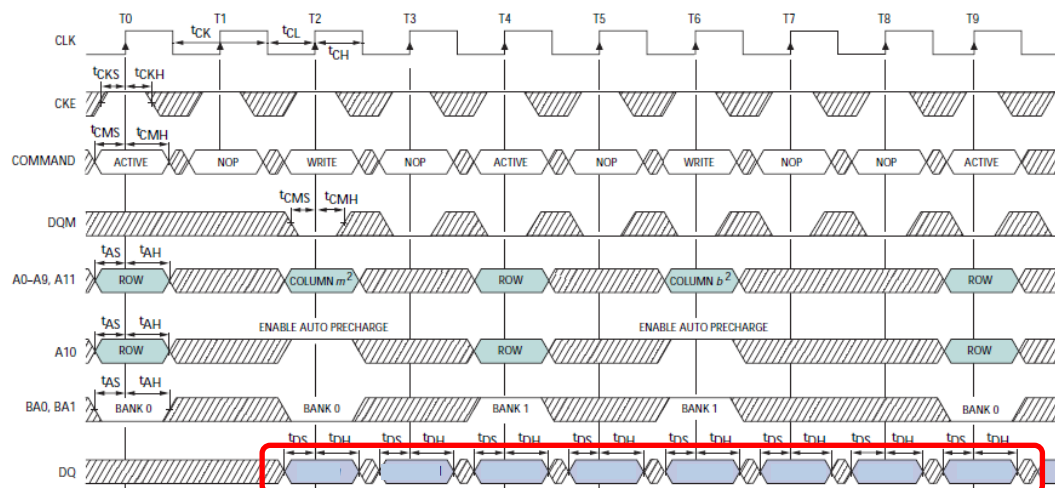
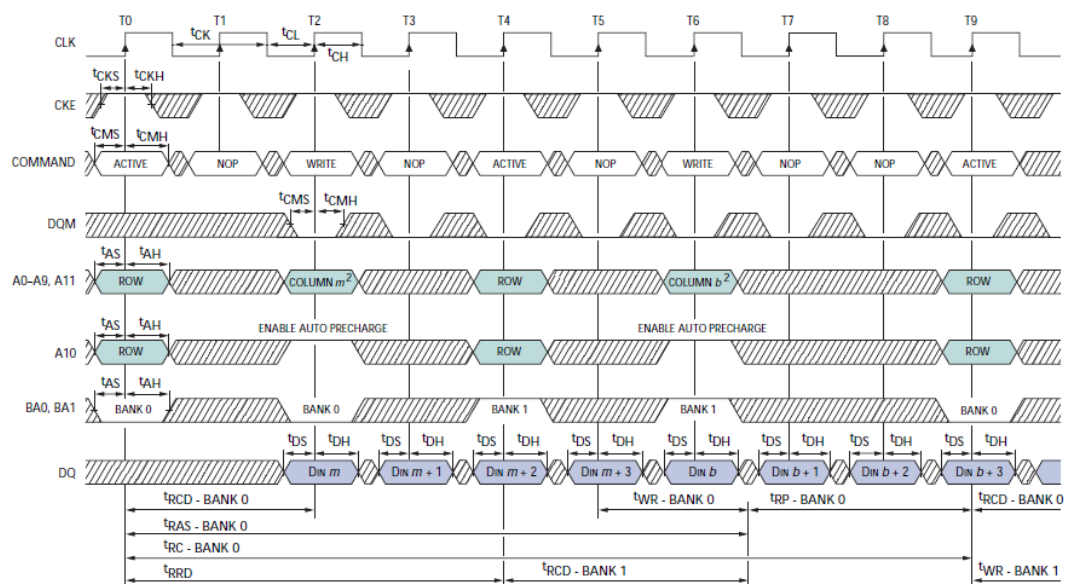


Figure 4.2: SDRAM Write – Alternating Bank Access (with Auto-Precharge)



5. What is the difference between auto-refresh and self-refresh operation? Reference Figure 5.1 below, t_{REF} parameter for this SDRAM, which has 8192 rows, is 64ms max. What would be the expected t_{REF} value for a SDRAM with 4096 rows? Assuming same SDRAM design and technology is used.

- Content of SDRAM, like DRAM, are stored in capacitor whose electrical charges tend to leak to the ground after some time, resulting in corrupted data. Refresh operation rewrite the original value of the cell content to preserve the data integrity. Auto-Refresh operation refreshes one row of a SDRAM each time the command is executed. Self-Refresh operation refreshes each row sequentially and continuously without further commands from the processor.
- t_{REF} is the maximum time required to refresh all the rows in the SDRAM. Time required to refresh all the rows remains unchanged if same SDRAM design and technology is use since the rate of leakage will be the same and all capacitors leaked at the same time! Hence, $t_{REF}=64ms$.

(Optional)

6. SSD Design considerations: homework given during lecture.

Over-provision

- over-provisioning refers to a function that secures extra space to allow for efficient use of the SSD by allocating a certain amount of the SSD's NAND flash to an over-provisioning space. This space can only be accessed by the SSD's controller and not by the host. Consisting of free blocks only, the OP region assists in efficient delivery of free blocks when wear-leveling or garbage collection is in progress and contributes to improved performance and lifetime of the SSD.
- Although there is no difference between the sequential and random write performance for fresh-out-of-the-box (FOB) NAND, the random write does not perform as well as the sequential write once data has been written over the entire space of the NAND. Random writes, smaller in size than sequential writes, mix valid and invalid pages within blocks, which causes frequent GC and results in decreased performance. If the OP is increased, more free space that is inaccessible by the host can be secured, and the resulting efficiency of GC contributes to improved performance. The sustained performance is improved in the same manner.

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SSD Design Consideration

- **Garbage Collection**
- Technique used to free up previously written blocks
- Consolidates pages by moving and rewriting pages from multiple blocks to fill up fewer new ones.
- Old blocks are then erased to provide storage space for new incoming data.

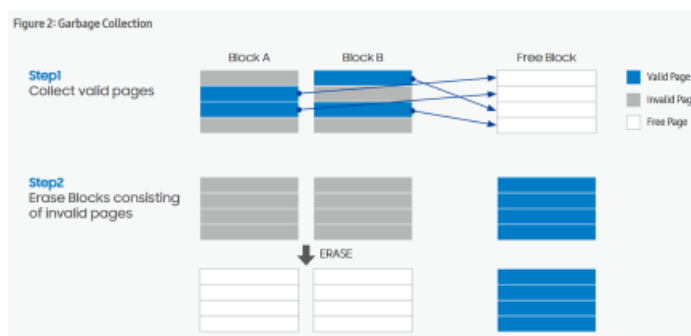


Image Source: Samsung

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SSD Design Consideration

- **Write Amplification**

- Smallest unit that can be programmed is a Page
- Smallest unit that can be erased is a Block (Multiple Pages)
- To overwrite a location in Flash, it has to be erased first (unlike HDD)
- As the drive is used, data changes, and the changed data is written to other pages in the block or to new blocks. At this point, the old (stale) pages are marked as invalid and can be reclaimed by erasing the entire block. To do this, however, any still-valid information on all of the other occupied pages in the block must be moved to another block.
- The requirement to relocate valid data and then erase blocks before writing new data into the same block causes **write amplification**;
- Write Amplification happens when the total number of writes required at the flash memory is higher than the host computer originally requested.
- It also causes the SSD to perform write operations at a slower rate when it is busy moving data from blocks that need to be erased while concurrently writing new data from the host computer.

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SSD Design Consideration

- **TRIM/UNMAP**

- The File Allocation Table maintained by the OS is not visible to the SSD, as such, when a file is deleted, the SSD do not know that the corresponding flash blocks can be freed. Only the storage corresponding to the FAT is modified.
- Certain Operating Systems support the TRIM/UNMAP function, which translates deleted files to the associated LBA (logical block address) on the storage device (SSD).
- The TRIM/UNMAP command notifies the drive it no longer needs data in certain LBAs (Logical Block Address) which then free up a number of NAND pages.
- The TRIM/UNMAP command needs to be supported by the OS, the drive, and the controller in order to work.
- The TRIM/UNMAP command could result in higher SSD performance from both the reduced data needed to be rewritten during garbage collection and the higher free space resulting on the drive.

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SSD Design Consideration

- When Host OS delete a file, only the FAT is modified, The actual data is still intact
- The Flash Controller does not know that the deleted locations are 'invalid' from Host OS point of view.
- The controller will continue to perform Garbage Collection for these locations. Meaning unnecessary data movement which reduces reliability, endurance and write performance.
- **TRIM/UMAP** command can be sent by OS to the Flash Controller to inform it of blocks that are marked invalid so that these block can be excluded from the Garbage collection and assigned to the free pool.